Design overview

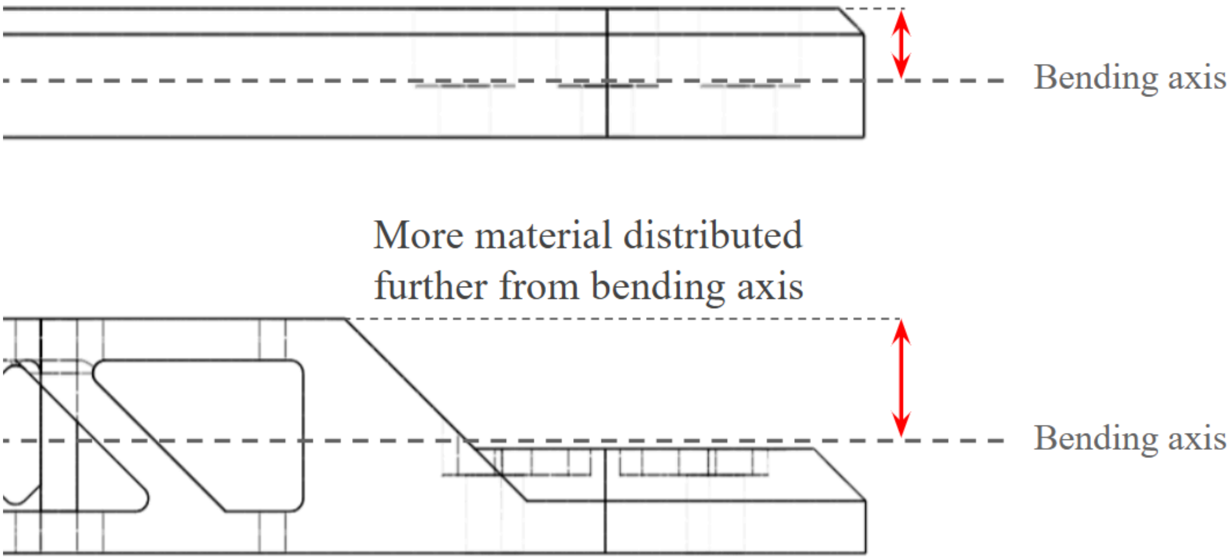
Frame stiffness improvement

The original frame design was designed to be manufactured out of carbon fibre which is solid and has a Young's modulus 20-30x higher than PLA plastic, which explains why the 3D printed frame shows significant flexure with applied load.

Bending stiffness is proportional to the **second moment of area** I. For a rectangular section, the second moment of area is given by:

$$I = \frac{bt^3}{12}$$

Where b is the width and t is the thickness in the bending direction. From the equation, we see that the bending stiffness scales the most with thickness t. The bending load is bore mainly by the fibres on the extreme ends in tension/compression; material close to the bending axis contributes exponentially less to the stiffness. Hence, bending stiffness is increased by re-distributing material further from the axis and using thin triangluar trusses to maintain structure.

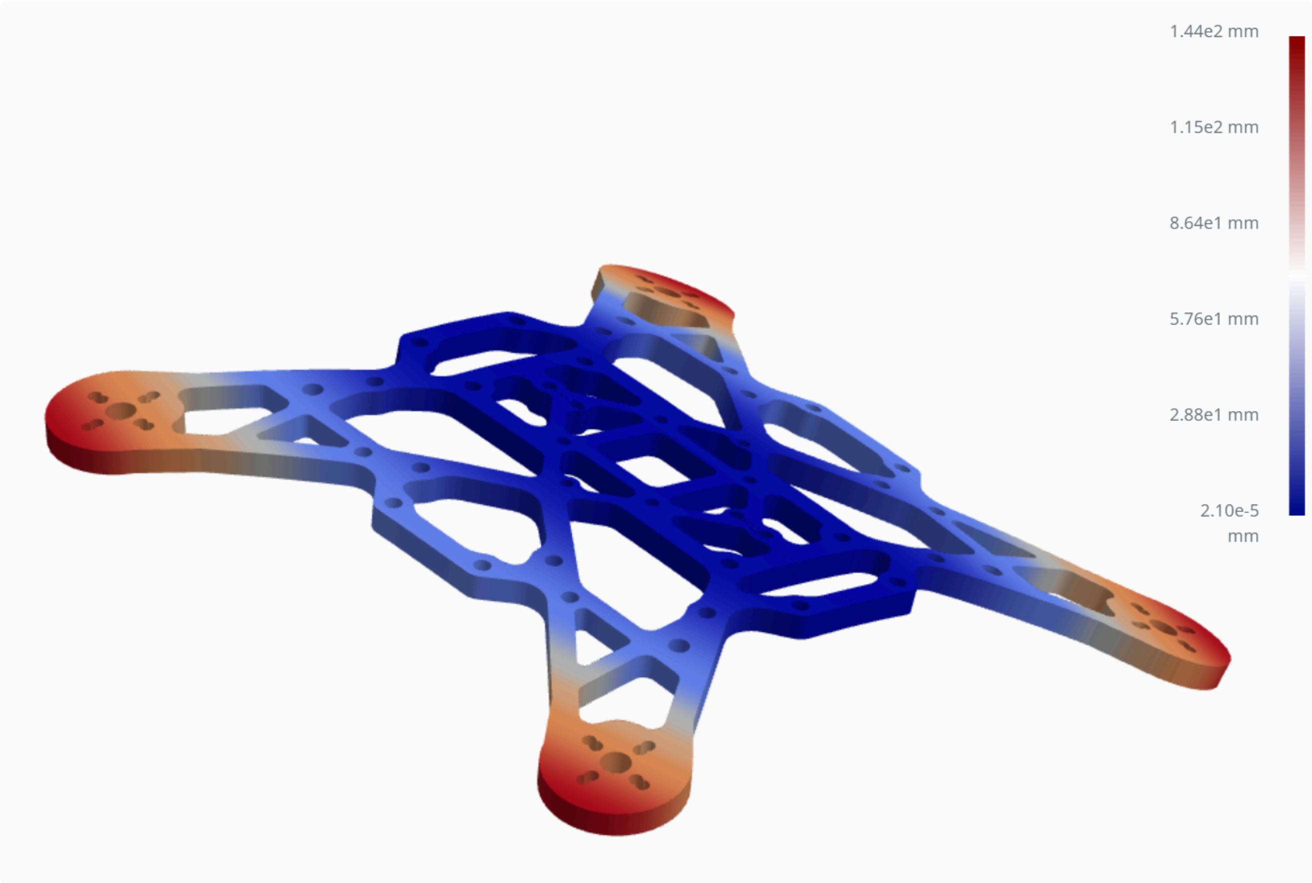


Analysis

A simple bending simulation was performed using [an online mesh simulator](). The model is fixed about the center and opposing bending moments were applied to either sides. The vertical deflection was recorded in millimeters.

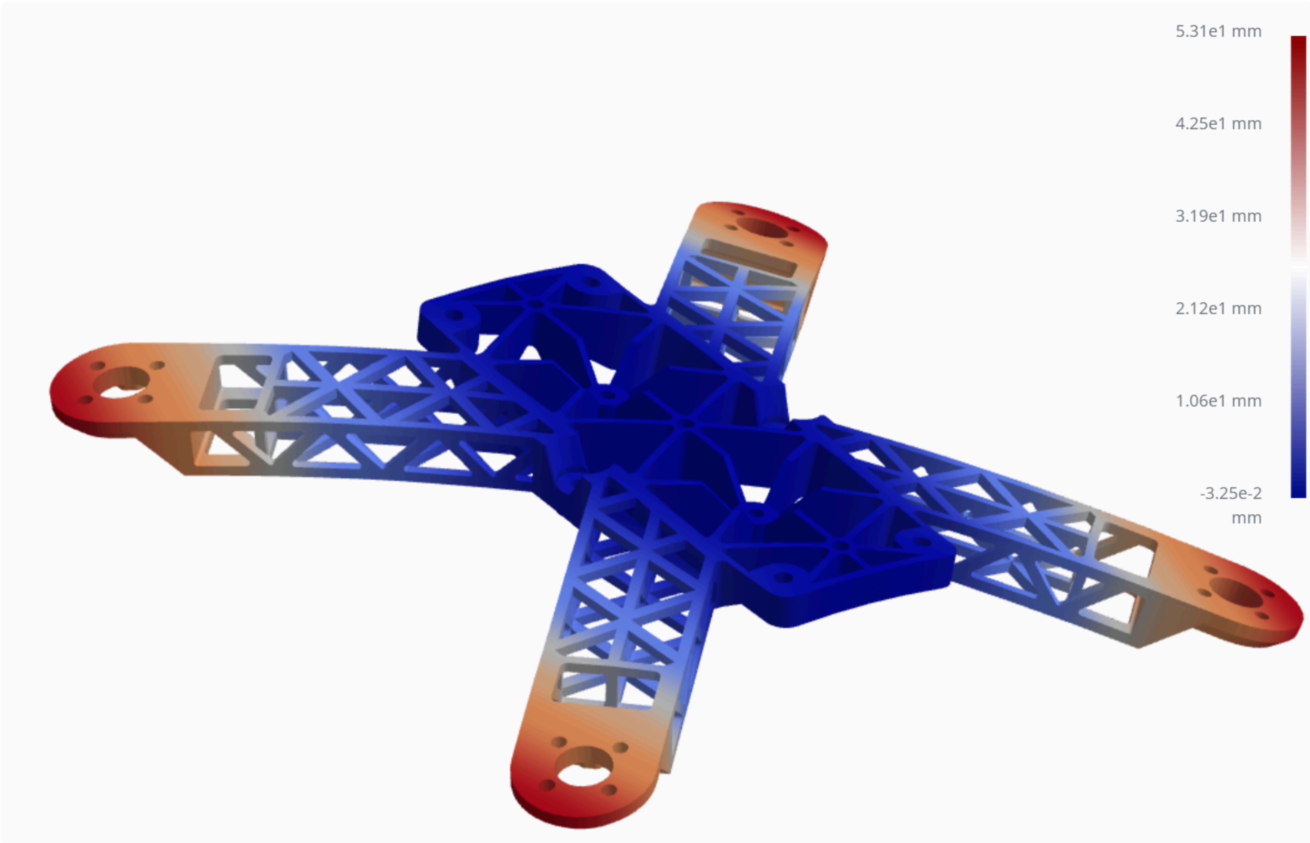
Parameter	Value
Bending moment	30N.m
Material	PLA

Original frame



Max vertical deflection: **14.4mm**

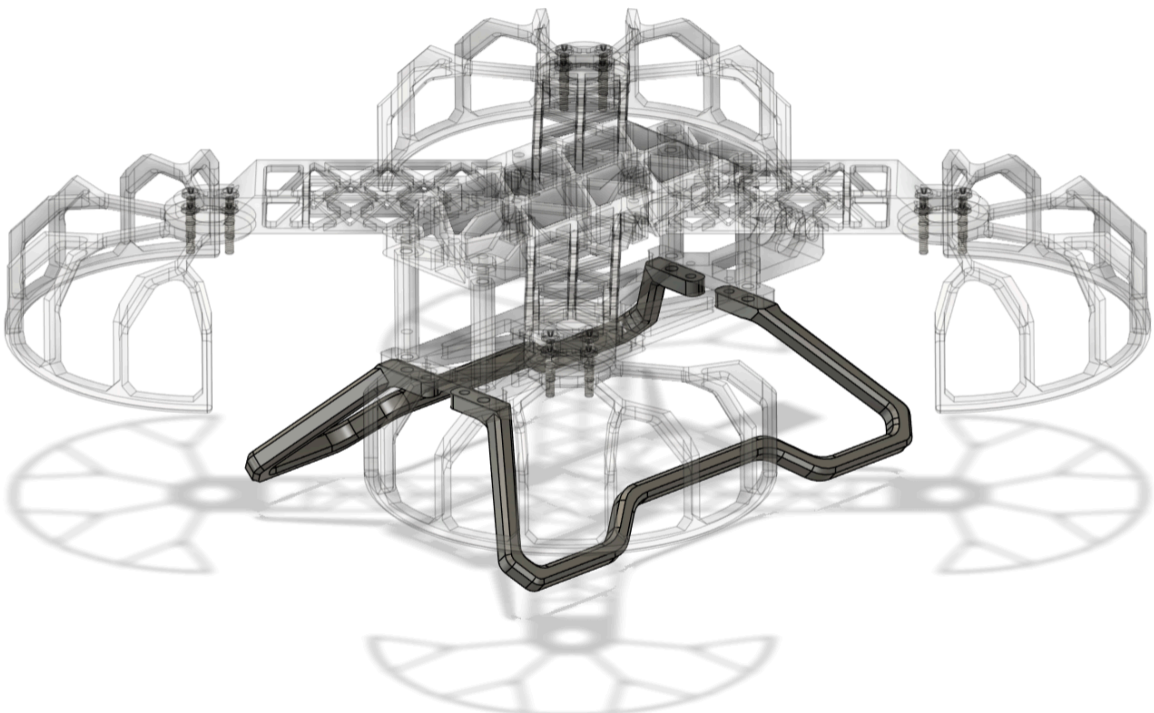
New frame



Max vertical deflection: **5.31mm**

The new frame is much stiffer and deflects $\approx 63\%$ less for the same bending moment applied.

Landing gear



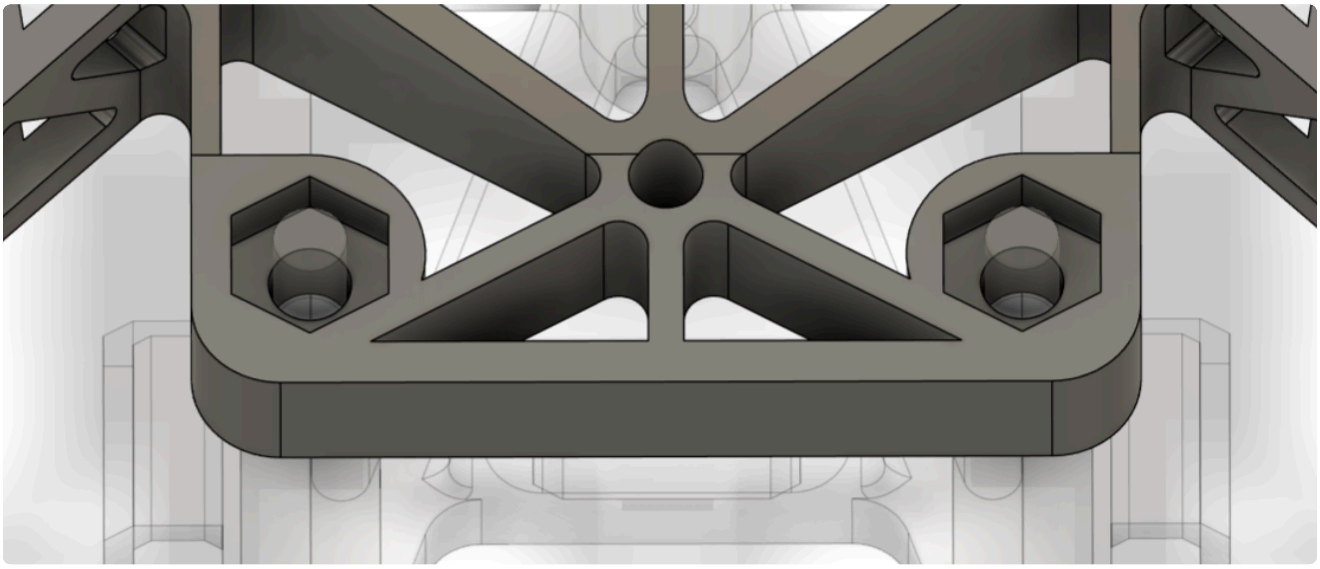
A set of easily replaceable landing feet is designed to splay outwards and absorb landing impact, replacing the nylon spacers in the original design which often came loose on impact.

Assembly improvements

Design features to make assembly more seamless. Original design takes ~1hr to assemble on average, new design ~15min.

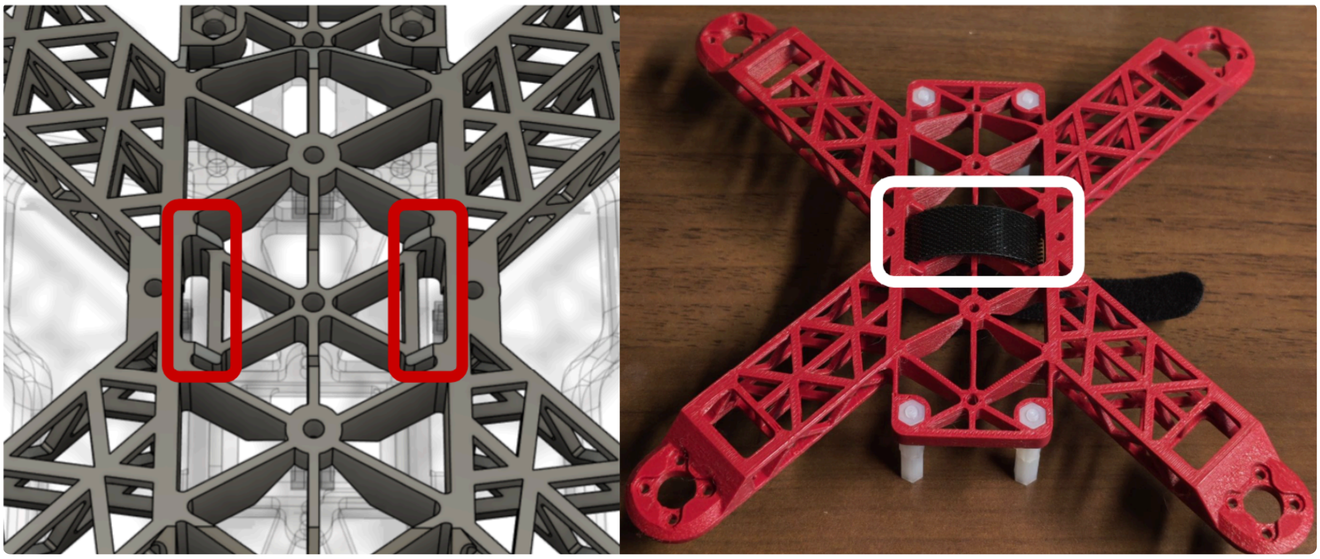
Nylon nut retainers

Mating nuts pressed into frame, tool-free installation of nylon stand-offs.



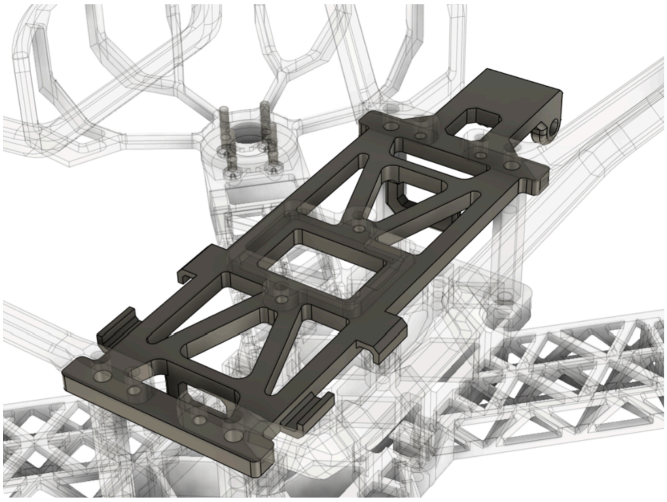
Integrated battery holder velcro passthrough

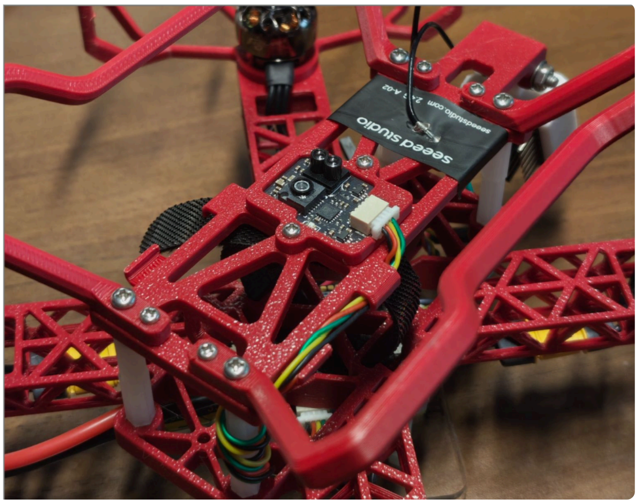
Recessed area with pass-through underneath flight controller to easily attach battery velcro strap.

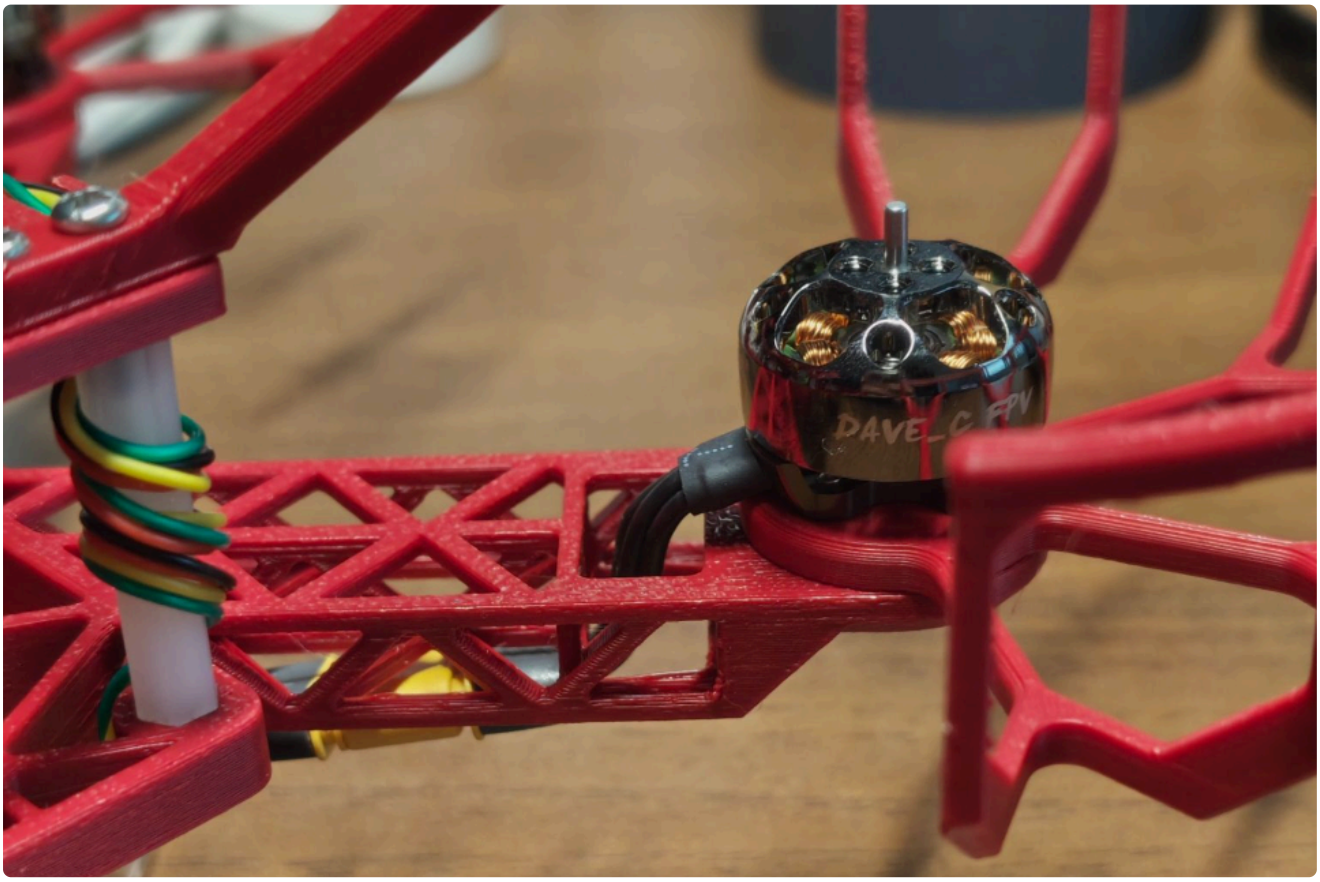


Wire channels

Modelled-in wire management channels for long connectors, clearing loose wires from propeller.

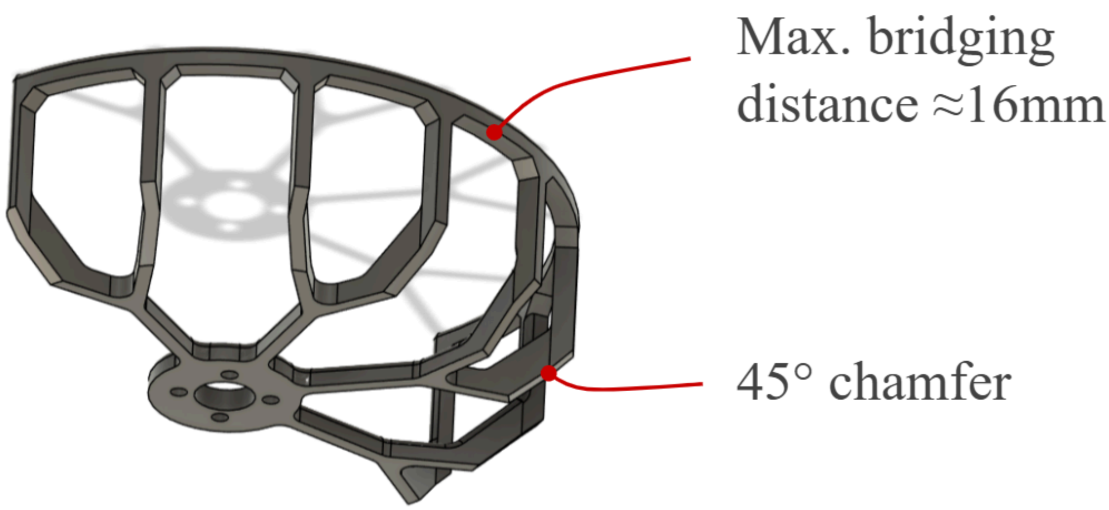


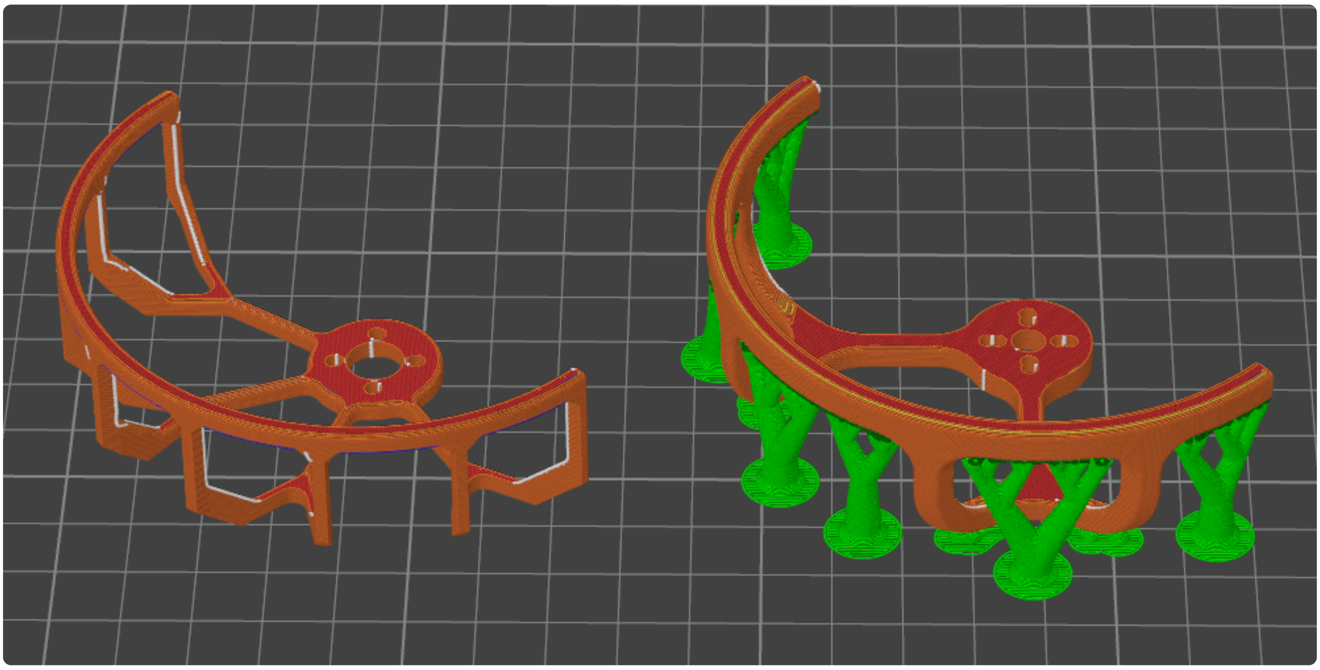




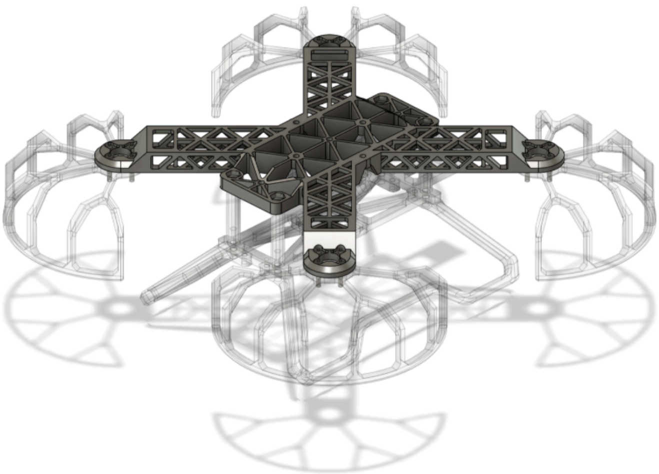
3D printing optimizations

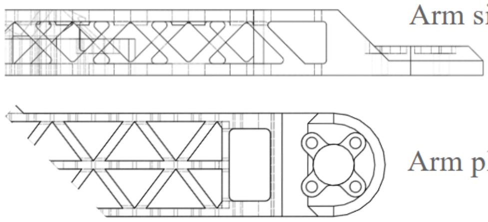
Common FDM 3D printers can comfortably bridge gaps of $\approx 20mm$ and print overhangs $< 55^{\circ}$ without droop when printing PLA. All components are designed to be printed without supports by exploiting these features.





New design (left) vs old (right): all supporting material eliminated by controlling bridging distance and overhang angle, saving time and material.

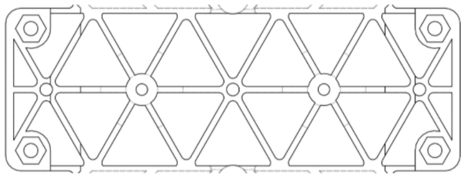




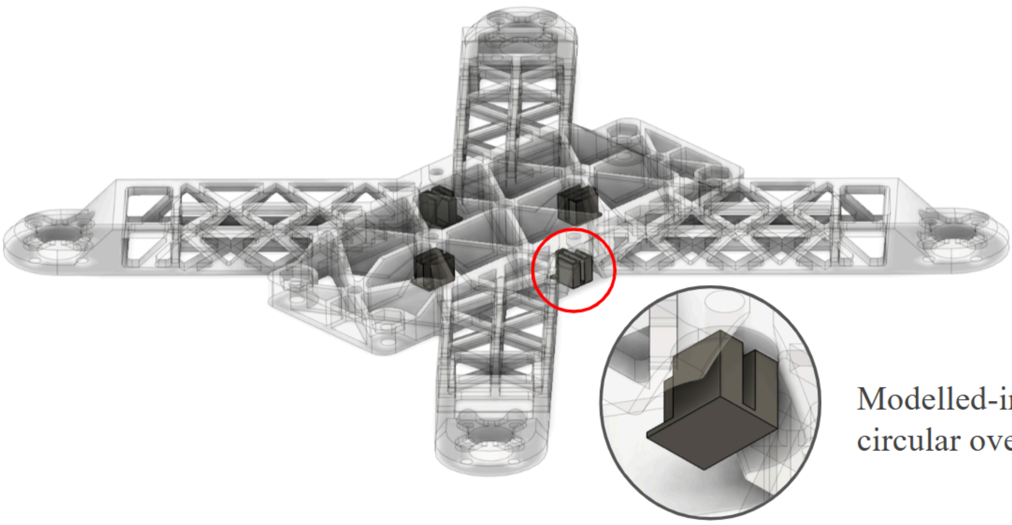
Arm side view

Arm plan view

Staggered triangles to support top lattice bridging.



Isogrid structure maintains strength while minimizing material.



Modelled-in supports for circular overhangs

Circular overhangs still require supports since the printer will try to print circles in air instead of bridging the gap. In such cases, supporting structures are directly modelled into the design.

Print time & weight comparison

The new frame design takes the same amount of time (≈ 3.5 hrs) to print as the original, and uses less filament since no supports are required. The total weight of the printed frame components is also similar to the original design at $\approx 50g$.

All frame components fit on a single standard $256 \times 256mm$ print bed.

The final drone assembly with all electrical components fitted (inlcuding battery) is around $224g$.